

Carrier Performance & SLA Risk Intelligence – Week 1 Case Study

name :Mostafa sharara

1. Business Problem Understanding

Apex Manufacturing Ltd. operates a complex B2B supply chain across APAC, EMEA, and AMER regions. Shipments are handled by multiple third-party carriers under strict Service Level Agreements (SLAs).

Challenges:

- SLA breaches identified after deliveries fail
- Carrier underperformance discovered too late
- Procurement relies on historical averages instead of predictive insights

Core Question:

Which carriers are likely to breach SLAs on specific routes, and why?

Objective:

Build a predictive, explainable platform to assess carrier performance, forecast SLA breaches, and enable proactive operational and procurement decisions.

2. Dataset Overview

File: `b2b_carrier_sla_risk_intelligence_100k.csv`

Rows: 200,000 shipments

Granularity: One row = one shipment

Key Columns & Business Meaning:

Column	Type	Business Meaning
shipment_id	String	Unique shipment identifier
vendor	Categorical	Carrier responsible for shipment
shipping_mode	Categorical	Road / Sea / Air / Rail

region	Categorical	APAC / EMEA / AMER
origin_country / destination_country	Categorical	Route definition
weight_kg / volume_cbm	Float	Shipment cost & capacity
priority_flag / fragile_flag	Binary	SLA strictness & handling risk
planned_delivery_days	Integer	SLA commitment
actual_delivery_days	Integer	Execution result
delivery_delay_days	Integer	SLA breach indicator
shipping_cost_usd	Float	Financial impact
sla_breach_flag	Binary	Target variable (1 = breach, 0 = on-time)

3. Data Issues Identified

- Missing values in numeric columns
 - Invalid values: `planned_delivery_days <= 0`, negative `shipping_cost_usd`
 - Outliers: extreme values in shipping cost, weight, volume, delivery delays
 - Categorical columns need encoding for ML
 - Target variable `sla_breach_flag` needed verification
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4. Cleaning Decisions & Justifications

1. **Missing Values:** Filled or removed based on criticality
2. **Invalid Values:** Removed negative shipping costs, verified planned delivery days > 0
3. **Outlier Handling:**
 - Cap shipment weight & volume using IQR
 - Keep extreme shipping costs and delivery delays (legitimate operational or premium shipments)

4. Categorical Encoding:

- Label Encoding: `vendor`, `origin_country`, `destination_country`
- One-Hot Encoding: `shipping_mode`, `region`
- Original categorical columns removed to avoid confusion

5. Target Variable Verification:

- `sla_breach_flag = 1 if actual_delivery_days > planned_delivery_days else 0`
- Verified across all rows

Business Rationale: All decisions aligned with operational KPIs and real-world logistics constraints.

5. Target Variable Logic

- **Binary:** 1 = SLA breached, 0 = SLA met
 - **Business Justification:** Any delay beyond SLA represents a contract breach. This is measurable and actionable.
 - **Class Distribution:** Imbalanced (most shipments on-time), which is expected in logistics.
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6. Final Dataset Summary

- All columns cleaned, encoded, and validated
- Outliers handled according to business rules
- Binary flags verified
- Dataset ready for ML: `week1_cleaned_data.csv`

Deliverables:

- Cleaned dataset: `week1_cleaned_data.csv`
 - Python notebook with structured, well-commented code
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7. Business Value

- Provides a trustworthy foundation for predictive SLA breach modeling
- Enables proactive carrier monitoring and risk mitigation
- Supports operational and procurement decisions
- Fully auditable and aligned with enterprise SLA KPI